

Muhammad Alif Dhiyaumasrur

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SUMMARY

Dedicated Mechatronics and Artificial Intelligence engineering student with strong expertise in edge artificial intelligence, computer vision, and industrial automation systems. Proven track record in building end to end smart hardware prototypes, leading high performing technical teams to national finalist stages, and mentoring student developers. Passionate about leveraging practical technology to create impactful human centric solutions.

EDUCATION

Universitas Pendidikan Indonesia

Purwakarta, Indonesia

Bachelor of Science in Mechatronics and Artificial Intelligence

Expected 2028

- **Coursework Focus:** Machine Learning, Deep Learning, Computer Vision, Internet of Things, Embedded Systems, Robotics, Programmable Logic Controllers (PLC), Industrial Automation.

EXPERIENCE & LEADERSHIP

Google Arcade Facilitator

Jul 2026 – Present

Google Cloud Developer Program

Purwakarta, Indonesia

- Mentored 50+ students through hands-on cloud computing and AI labs, fostering a campus-wide developer community.
- Guided peers in troubleshooting technical issues and maintaining engagement in emerging technology ecosystems.

Machine Learning Engineering Intern

Jul 2026 – Aug 2026

PT. Astra Honda Motor (AHM)

Indonesia

- Developed an end-to-end predictive maintenance system for industrial metrology equipment (CMM and Contour).
- Built ETL pipelines to parse MTConnect XML streams, storing data in InfluxDB and PostgreSQL.
- Trained anomaly detection and Remaining Useful Life (RUL) models using a XGBoost, build Auto-Work Orders and visualized on a Streamlit dashboard.

Substitute Lecturer / Teaching Assistant

2026

Universitas Pendidikan Indonesia

Purwakarta, Indonesia

- Delivered guest lectures and lab sessions across multiple year cohorts covering foundational and advanced computer science topics.
- Taught Computational Thinking for the 2026 cohort and Object-Oriented Programming (Java Encapsulation) for the 2025 cohort.
- Instructed peers in the 2024 cohort on Introduction to Machine Learning, focusing on the necessity of ML and Exploratory Data Analysis (EDA).

PROJECTS

TANGGAP-AI (Offline Emergency Response LLM) | *Gemma 4 E2B, LiteRT-LM, RAG*

May 2026

- Architected an offline-capable Android application for emergency scenarios using local LLMs (Gemma 4 E2B via LiteRT-LM).
- Implemented a BM25 RAG (Retrieval-Augmented Generation) pipeline to provide accurate context-aware responses locally.
- Developed as part of the Google Gemma 4 Hackathon.

Monetra (Visual Currency Reader) | *YOLOv8, ESP32-CAM, Edge AI*

Aug 2025 – Feb 2026

- Built an embedded AI hardware prototype to assist visually impaired individuals in identifying currency.
- Deployed YOLOv8 object detection models onto an ESP32-CAM microcontroller for edge AI inference.
- Achieved Top 10 National Finalist in Samsung Innovation Campus Batch 7.

Transformer Architecture Implementation | *PyTorch, Python*

2025

- Programmed a complete PyTorch implementation of the Transformer model from scratch based on the "Attention Is All You Need" paper to deepen understanding of LLM architectures.

- Legal RAG Chatbot (Fine-tuned Llama-3-8B + GRPO)** | *Unsloth, TRL, ChromaDB, RAG* 2026
- Developed an Indonesian legal chatbot by fine-tuning Llama-3-8B via LoRA and training reasoning skills using GRPO reinforcement learning.
 - Architected an Advanced RAG pipeline featuring hybrid search (BM25 + Vector), HyDE, and Cross-Encoder reranking.
 - Published the fine-tuned model publicly on Hugging Face ([Saranjana/Llama-3-8b-Legal-RAG](#)).
- Bitcoin Price Forecasting with Deep Learning** | *TensorFlow, LSTM, Custom Attention, Python* 2026
- Engineered a time-series forecasting pipeline to predict 24-hour Bitcoin closing prices using hourly historical data.
 - Built custom layers and mechanisms from scratch using TensorFlow (Dense, Multi-Head Attention, LeakyReLU).
 - Implemented a custom training loop via `tf.GradientTape` with a weighted MAE loss function, outperforming a Seq2Seq baseline model.
- MAPAN (AI Career Transition Platform)** | *NER, Web Development* Apr 2026 – Jun 2026
- Designed a platform featuring automated CV parsing using Named Entity Recognition (NER) to analyze skill gaps.

AWARDS & CERTIFICATIONS

- **Gold Medal & Runner-Up:** Edutalk Fair Competition 2026 at Universitas Diponegoro for the "AyoBikin" project.
- **Top 10 National Finalist:** Samsung Innovation Campus Batch 7 (for the "Monetra" AI Embedded Project, 2026).
- **Top 20 National Finalist:** Nestlé - #GreenGeneration Sustainability Business Case Competition (2026).
- **NVIDIA Deep Learning Institute:** Certified in Fundamentals of Deep Learning & Building Transformer-Based NLP Applications (2026).
- **IDCamp 2025 (Indosat & Dicoding):** Graduated as Gen AI Engineer up to Level Expert (Final Tier).
- **DBS Foundation Coding Camp 2025:** Certified AI Engineer Learning Path (Dicoding).
- **Microsoft Elevate Training Center:** Completed all training stages to the final level.

TECHNICAL SKILLS

- **Languages:** Python, C/C++, Java, SQL, JavaScript (Node.js, React), HTML/CSS
- **AI/ML Frameworks:** PyTorch, TensorFlow, YOLOv8, XGBoost, LiteRT, RAG Pipelines, Ollama
- **Hardware & IoT:** ESP32, Raspberry Pi, Arduino, Omron PLC (CX-Programmer), AutoCAD, Bambu Studio
- **Tools & Environment:** Docker, Linux, Git, MLflow, VSCode, OpenCode